2/4/2026

HW: 4. Solve $(D^2-3D+2)y = 2e^{x}\cos x/2$ [Ans: $c_1e^{2x}+c_2e^{x}+\frac{8e^{x}}{5}\left[-2\sin\frac{x}{2}-\cos\frac{x}{2}\right]$

Eg 3: Solve $(D^2+4D+4)y = \dfrac{e^{-2x}}{x^5}$

Soln: Auxiliary eqn is:

$$D^2+4D+4=0$$

$$\Rightarrow D=-2,-2$$

$\therefore$ C.F is:

$$y_c = (c_1+c_2x)e^{-2x}$$

P.I is:

$$y_p = \frac{1}{(D^2+4D+4)}\frac{e^{-2x}}{x^5}$$

Rule

$$\frac{1}{f(D)}e^{ax}P(x) = e^{ax}\frac{1}{f(D+a)}P(x)$$

$$\Rightarrow y_p = e^{-2x}\left\{\frac{1}{(D-2)^2+4(D-2)+4}\left(\frac{1}{x^5}\right)\right\}$$

$a=-2$

$$= e^{-2x}\left\{\frac{1}{D^2-\cancel{4D}+\cancel{4}+\cancel{4D}-\cancel{8}+\cancel{4}}\right\}x^{-5}$$

$$\Rightarrow y_p = e^{-2x}\cdot\frac{1}{D^2}x^{-5}$$

$$= e^{-2x}\cdot\frac{1}{D}\left[\frac{1}{D}x^{-5}\right]$$

$$= e^{-2x}\cdot\int\left[\int x^{-5}dx\right]dx$$

$$\Rightarrow y_p = e^{-2x}\cdot\int\frac{x^{-4}}{-4}dx$$

ie $y_p = e^{-2x}\cdot\dfrac{x^{-3}}{(-3)(-4)} = \dfrac{e^{-2x}\cdot x^{-3}}{12}$

$\therefore$ General soln is:

$$y = y_c + y_p$$

Method of Variation of Parameters

Consider a <u>second order linear d.e</u>:

$$f(D)y = Q(x).$$

Let the C.F be:

$$y_c = c_1 y_1(x) + c_2 y_2(x)$$

Then the P.I will be

$$y_p = u(x)\, y_1(x) + v(x)\, y_2(x)$$

Where $u(x) = -\int \frac{y_2(x) \cdot Q(x)}{W}\, dx$

and $v(x) = \int \frac{y_1(x)\, Q(x)}{W}\, dx$

Where $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$ is the <u>Wronksian</u>

$y_1' = \frac{dy_1}{dx}$

Egs

1. Solve by MVP: $(D^2-4D+4)y = e^{2x}\sec^2 x$

Sol^n: Aux eq^n is:

$$D^2-4D+4 = 0$$

$$\Rightarrow D = 2, 2$$

$\therefore$ C.F: $y_c = (c_1 + c_2 x)e^{2x}$

ie $y_c = c_1 \underbrace{e^{2x}}_{y_1(x)} + c_2 \underbrace{x \cdot e^{2x}}_{y_2}$

Let $y_1(x) = e^{2x}$; $y_2(x) = x e^{2x}$

Let the P.I

$$y_p = u(x)\, y_1(x) + v(x)\, y_2(x).$$

Where $u(x) = -\int \frac{y_2(x)\, Q(x)}{W}\, dx$

and $v(x) = \int \frac{y_1(x)\, Q(x)}{W}\, dx$

Here $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$

$$\Rightarrow W = \begin{vmatrix} e^{2x} & x \cdot e^{2x} \\ 2e^{2x} & 2xe^{2x} + e^{2x} \end{vmatrix}$$

$$\Rightarrow W = e^{2x}(2xe^{2x} + e^{2x}) - xe^{2x} \cdot 2e^{2x}$$

$$\Rightarrow W = 2x\cancel{e^{4x}} + e^{4x} - 2x\cancel{e^{4x}}$$

$$\Rightarrow \underline{\underline{W = e^{4x}}}$$

$$\therefore u(x) = -\int \frac{x\cancel{e^{2x}} \cdot \cancel{e^{2x}} \cdot \sec^2 x}{\cancel{e^{4x}}}\, dx$$

$$\Rightarrow u(x) = -\int x \sec^2 x\, dx$$

$$= -\left[(x)\tan x - (1)\log \sec x\right]$$

$$= -x\tan x + \log \sec x$$

& $v(x) = \int \frac{\cancel{e^{2x}} \cdot \cancel{e^{2x}} \sec^2 x}{\cancel{e^{4x}}}\, dx$

$$\Rightarrow v(x) = \int \sec^2 x\, dx = \tan x.$$

$$\therefore y_p = (-x\tan x + \log \sec x)e^{2x} + \tan x \cdot x e^{2x}$$

& gen soln: $y = y_c + y_p$.

2. Solve $\frac{d^2y}{dx^2}+y=\frac{1}{1+\sin x}$

Sol$^{\underline{n}}$: Let $D=\frac{d}{dx}$

$\therefore$ the given d.e is:

$$(D^2+1)y=\frac{1}{1+\sin x}$$

Aux eq$^{\underline{n}}$ is:

$$D^2+1=0 \Rightarrow D=\pm i$$

$\therefore$ C.F: $y_c=e^{0.x}(c_1\cos x+c_2\sin x)$

ie $y_c=c_1\cos x+c_2\sin x$

Let $y_1(x)=\cos x$; $y_2(x)=\sin x$

Let P.I be

$$y_p=u(x)\cdot y_1(x)+v(x)\,y_2(x)$$

ie $y_p=u(x)\cdot\cos x+v(x)\sin x$

Now

$$W=\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}=\begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}$$

$\Rightarrow W=\cos^2 x+\sin^2 x=1$

& we have

$$Q(x)=\frac{1}{1+\sin x}$$

$$\therefore u(x)=-\int\frac{y_2(x)Q(x)}{W}dx \;\&\; v(x)=\int\frac{y_1(x)Q(x)}{W}dx$$

$$\Rightarrow u(x)=-\int\frac{\sin x}{1+\sin x}dx$$

$$=-\int\frac{(\sin x)(1-\sin x)}{(1+\sin x)(1-\sin x)}dx$$

$$=-\int\frac{\sin x-\sin^2 x}{1-\sin^2 x}dx$$

$$=-\int\left(\frac{\sin x}{\cos^2 x}-\frac{\sin^2 x}{\cos^2 x}\right)dx$$

$$=-\int\left(\frac{\sin x}{\cos x}\cdot\frac{1}{\cos x}-\tan^2 x\right)dx$$

$$\Rightarrow u(x)=-\int\left[\tan x\sec x-(\sec^2 x-1)\right]dx$$

$$\Rightarrow u(x)=-\int\tan x\sec x\,dx+\int\sec^2 x\,dx-\int dx$$

$$\frac{\sin x}{1+\sin x} = \frac{\sin x\,(1-\sin x)}{(1+\sin x)(1-\sin x)}$$

or

$$\frac{2\sin x/2\,\cos x/2}{\cos^2 x/2 + \sin^2 x/2 + 2\sin x/2\cos x/2}$$

$$Dr = \left(\cos x/2 + \sin x/2\right)^2$$

$$\Rightarrow u(x) = -\int \sec x \tan x\,dx + \int \sec^2 x\,dx - \int dx$$

$$\Rightarrow u(x) = -\sec x + \tan x - x$$

$$\& \; v(x) = \int \frac{y_1(x)\,Q(x)}{W}\,dx = \int \frac{\cos x \cdot 1}{1+\sin x}\,dx$$

put $1+\sin x = t$

$\Rightarrow \cos x\,dx = dt$

$$\therefore v(x) = \int \frac{dt}{t} = \log t = \log(1+\sin x)$$

$$\therefore y_p = u(x)\,y_1(x) + v(x)\,y_2(x)$$

$$= (-\sec x + \tan x - x)\cos x + \log(1+\sin x)\sin x$$

& gen. solⁿ is: $y = y_c + y_p$